BANK LOAN PORTFOLIO ANALYSIS

1. Project Overview

The Bank Loan Portfolio Analysis project focuses on analyzing loan data to understand lending performance, customer borrowing patterns, and loan repayment trends. The dataset contains 38,576 loan records.

In this project, I used Python for data cleaning, PostgreSQL and SQL for data analysis, and Power BI to create an interactive dashboard.

The dashboard shows important metrics such as total loan applications, total funded amount, total amount received, average interest rate, and average debt-to-income ratio. It also helps identify trends in loan performance and potential credit risks.

2. Project Objectives

The main objectives of this project are:

- Analyze 38,576 loan records to understand overall lending performance.
- Calculate key metrics such as total loan applications, funded amount, amount received, average interest rate, and debt-to-income ratio.
- Analyze loan repayment status to identify fully paid, charged-off, and current loans.
- Understand loan distribution based on grade, purpose, income category, employment length, and homeownership.
- Identify potential credit risks by analyzing charge-off rates across different loan grades.
- Analyze monthly loan application trends to understand changes in lending activity.
- Build an interactive Power BI dashboard to present the findings clearly.

3. Dataset Description

This project uses a dataset of 38,576 bank loan records. It includes information about loan applications, customer financial details, loan amounts, interest rates, repayment status, and other important lending factors.

The dataset contains 29 columns that help analyze loan performance and identify potential credit risks.

Important Columns in the Dataset:

- **Loan ID:** A unique identification number assigned to each loan.
- **Loan Amount:** The amount of money requested by the borrower.
- **Total Payment:** The total amount received toward the loan.
- **Interest Rate:** The interest rate associated with the loan.
- **DTI (Debt-to-Income Ratio):** A measure of the borrower's debt compared to their income.
- **Loan Status:** Indicates whether the loan is Fully Paid, Charged Off, or Current.
- **Loan Grade:** Represents the loan's credit grade, ranging from A to G.

- **Annual Income:** The borrower's reported yearly income.
- **Home Ownership:** Indicates whether the borrower rents, owns, or has a mortgage.
- **Employment Length:** The borrower's reported employment duration.
- **Loan Purpose:** The reason for taking the loan.
- **Issue Date:** The date on which the loan was issued.
- **Address State:** The borrower's state.
- **Income Category:** Groups borrowers based on income levels.
- **Risk Level:** Categorizes loans based on the risk classification available in the dataset.

Data Source and Storage:

The cleaned loan dataset was stored in a PostgreSQL database and connected to Power BI for analysis and visualization.

4. Tools and Technologies Used

The following tools and technologies were used to complete the Bank Loan Portfolio Analysis project.

- **Python:** Python was used for data cleaning and preprocessing. The Pandas library helped identify missing values, check data types, handle duplicate records, and prepare the dataset for analysis.
- **Jupyter Notebook:** Jupyter Notebook was used to write and execute Python code. It helped organize the data cleaning process and review the dataset before loading it into the database.
- **PostgreSQL:** PostgreSQL was used to store and manage the cleaned loan dataset containing 38,576 records.
- **SQL:** SQL was used to query and analyze loan data. Queries included calculations of total loan applications, funded amounts, repayment amounts, loan status distribution, and credit risk indicators.
- **pgAdmin 4:** pgAdmin 4 was used to connect to the PostgreSQL database, execute SQL queries, and review the results.
- **Power BI:** Power BI was used to create an interactive dashboard displaying important loan metrics, repayment performance, loan categories, credit risk, and monthly lending trends.
- **DAX (Data Analysis Expressions):** DAX measures were created in Power BI to calculate total loan applications, total funded amount, total amount received, average interest rate, average debt-to-income ratio, and charge-off rate.

5. Data Cleaning and Preprocessing

Data cleaning and preprocessing were important steps in preparing the bank loan dataset for analysis. Python and the Pandas library were used to examine the data and prepare it for further analysis.

- **Data Loading:** The loan dataset was loaded into Jupyter Notebook using Python and Pandas to understand its structure and review the available columns.
- **Data Quality Checks:** The dataset was examined for missing values, duplicate records, inconsistent data types, and other potential data quality issues.
- **Data Type Validation:** Important columns such as loan amount, interest rate, debt-to-income ratio, and loan issue date were reviewed to ensure they could be used correctly in calculations and visualizations.
- **Data Categorization:** The cleaned dataset included categories such as income category, loan amount category, DTI category, and risk level to support further analysis.
- **Data Preparation:** The dataset was prepared for analysis and saved as cleaned_financial_loan.csv.
- **Loading Data into PostgreSQL:** The cleaned dataset containing 38,576 loan records was loaded into PostgreSQL. SQL queries were then used to analyze lending performance, repayment patterns, and credit risk.
- **Power BI Data Transformation:** The PostgreSQL dataset was connected to Power BI. The loan issue date was converted to a date format, and a corrected DTI category column was created to classify loans into Low, Medium, and High DTI groups.

These steps helped prepare the dataset for SQL analysis and Power BI reporting.

6. SQL Data Analysis

SQL was used to analyze the cleaned bank loan dataset stored in PostgreSQL. The analysis focused on loan applications, funded amounts, repayment performance, customer characteristics, and credit risk.

- **Total Loan Applications:** The dataset contains 38,576 loan applications. SQL was used to calculate the total number of records and understand the overall loan portfolio size.
- **Total Funded Amount:** The total funded amount was approximately $436 million. This represents the total loan amount issued to borrowers.
- **Total Amount Received**: The total amount received was approximately $473 million. This represents the sum of recorded loan payments across the dataset.
- **Average Interest Rate**: The average interest rate across the loan portfolio was 12.05%.
- **Average Debt-to-Income Ratio**: The average debt-to-income (DTI) ratio was 13.33%. This metric helps understand borrowers' debt obligations relative to their income.
- **Loan Status Analysis**: SQL was used to group loans based on repayment status. The results were:

Loan Status	Applications	Percentage
Fully Paid	32,145	83.33%
Charged Off	5,333	13.82%
Current	1,098	2.85%
Total	**38,576**	**100%**

The majority of loans were fully paid, while 13.82% were charged off.

- **Loan Grade Analysis:** Loan applications were analyzed across credit grades A through G. Grade B had the highest number of applications, followed by Grades A and C.
- **Loan Purpose Analysis:** The analysis showed that debt consolidation was the most common reason for borrowing, followed by credit card repayment.
- **Home Ownership Analysis:** The largest borrower groups were renters and mortgage holders. The dataset contained 18,439 loans associated with renters and 17,198 with mortgage holders.
- **Credit Risk Analysis:** Charge-off rates were compared across loan grades. The analysis showed higher charge-off rates among lower credit grades, particularly Grades F and G.
- **Monthly Loan Application Trends:** Loan applications were grouped by issue month to understand lending activity over time. The analysis showed an overall increase in monthly applications during 2021.

Summary

The SQL analysis provided important insights into loan distribution, repayment performance, borrower characteristics, and credit risk. These findings were used to develop the Power BI dashboard.

7. Power BI Dashboard Development

Power BI was used to develop an interactive dashboard for analyzing bank loan portfolio performance. The cleaned dataset containing 38,576 loan records was connected to Power BI through PostgreSQL.

The dashboard was designed to present key financial metrics, loan application patterns, repayment performance, and credit risk in a clear and understandable format.

7.1 Key Performance Indicators (KPIs): Five KPI cards were created to display the overall performance of the loan portfolio.

KPI	Value
Total Loan Applications	38,576
Total Funded Amount	$436 Million
Total Amount Received	$473 Million
Average Interest Rate	12.05%
Average DTI	13.33%

These KPIs provide a quick overview of the bank's lending activity and portfolio performance.

7.2 DAX Measures: DAX (Data Analysis Expressions) was used to create measures for calculating important financial metrics.

The following measures were developed:

- **Total Loan Applications:** Counts the total number of loan records.

- **Total Funded Amount:** Calculates the sum of loan amounts.
- **Total Amount Received:** Calculates the total recorded loan payments.
- **Average Interest Rate:** Calculates the average interest rate across loans.
- **Average DTI:** Calculates the average debt-to-income ratio.
- **Charge-Off Rate:** Calculates the percentage of loans classified as charged off within the selected group.

7.3 Dashboard Visualizations: Ten charts were created to analyze different aspects of the loan portfolio.

- **Loan Status Distribution:** A donut chart showing Fully Paid, Charged Off, and Current loans.
- **Loan Applications by Grade:** A column chart showing loan applications across Grades A to G.
- **Loan Applications by Purpose:** A bar chart showing the most common borrowing purposes.
- **Loan Applications by Home Ownership:** A bar chart comparing loan applications by home ownership status.
- **Loan Applications by Income Category:** A column chart showing the distribution of applications across income groups.
- **Loan Applications by DTI Category:** A column chart classifying loans into Low, Medium, and High DTI groups using analysis-defined thresholds.
- **Loan Applications by Employment Length:** A bar chart comparing loan applications based on employment duration.
- **Loan Applications by State:** A bar chart showing loan application distribution across US states.
- **Monthly Loan Application Trends:** A line chart showing monthly lending activity during 2021.
- **Charge-Off Rate by Loan Grade:** A column chart comparing charge-off percentages across loan grades.

7.4 Dashboard Design: The dashboard was organized with a main title, five KPI cards, and ten visualizations. Clear chart titles, consistent formatting, and a structured layout were used to make the information easy to understand.

Power BI's interactive features allow users to select different chart categories and explore related loan information.

7.5 Dashboard Outcome: The completed dashboard provides a consolidated view of loan portfolio performance. It helps users understand borrowing patterns, monitor repayment status, compare loan grades, and identify areas of potential credit risk.

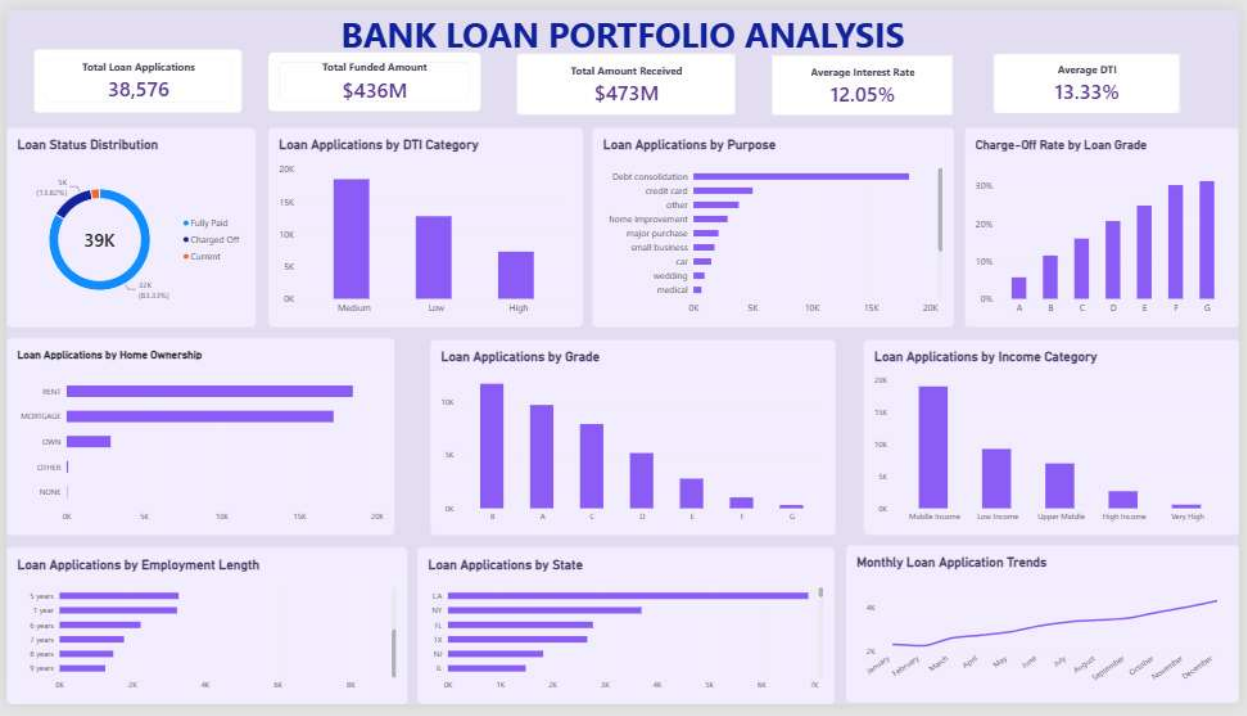


Figure 1: Bank Loan Portfolio Analysis Dashboard.

8. Key Findings and Business Insights: The Bank Loan Portfolio Analysis project provided important insights into lending performance, borrower characteristics, repayment patterns, and credit risk.

- **Overall Loan Portfolio Performance:** The dataset contains 38,576 loan applications, with approximately $436 million in total funded loans and $473 million in recorded payments. These figures provide an overview of the portfolio's lending and repayment activity.
- **Loan Repayment Performance:** Approximately 83.33% of loans were fully paid, while 13.82% were charged off and 2.85% were current. Although most loans were fully paid, the charged-off loans represent an important area for credit risk monitoring.
- **Loan Grade Distribution:** Grade B had the highest number of loan applications, followed by Grades A and C. This indicates that a large portion of the loan portfolio was concentrated in these credit grades.
- **Credit Risk by Loan Grade:** The charge-off rate generally increased from Grade A to Grade G. Lower credit grades, particularly F and G, showed higher charge-off rates, suggesting greater repayment risk in these groups.
- **Loan Purpose Analysis:** Debt consolidation was the most common loan purpose, followed by credit card repayment. This suggests that many borrowers used loans to manage or consolidate existing debt.
- **Home Ownership Analysis:** Renters accounted for 18,439 loan applications, while mortgage holders accounted for 17,198. These two groups represented the majority of borrowers in the dataset.

- **Debt-to-Income Ratio Analysis:** The average DTI was 13.33%. Borrowers were also grouped into Low, Medium, and High DTI categories to understand how debt obligations varied across the portfolio.
- **Monthly Loan Application Trends:** The monthly loan application chart showed an overall increase in lending activity during 2021, with December recording the highest number of applications.

Business Recommendations

Based on these findings, the following recommendations can be considered:

- Monitor loans in lower credit grades more closely because they have higher charge-off rates.
- Review lending policies and repayment patterns for borrowers with higher DTI ratios.
- Analyze debt consolidation loans to understand borrower needs and repayment behavior.
- Track monthly application trends to support lending forecasts and operational planning.
- Use Power BI dashboards to monitor portfolio performance and identify potential risks.

These findings can help financial institutions make more informed lending decisions and improve credit risk monitoring.

9. Conclusion

The Bank Loan Portfolio Analysis project successfully analyzed 38,576 loan records to understand lending performance, borrower characteristics, repayment patterns, and credit risk.

Python and Pandas were used for data cleaning and preprocessing, while PostgreSQL and SQL were used to store, manage, and analyze the dataset. Power BI was used to develop an interactive dashboard with key performance indicators and visualizations.

The analysis showed that 83.33% of loans were fully paid, while 13.82% were charged off. It also identified debt consolidation as the most common loan purpose and highlighted higher charge-off rates among lower credit grades.

The Power BI dashboard provided a clear overview of loan applications, funded amounts, repayments, interest rates, debt-to-income ratios, and monthly lending trends.

Overall, this project demonstrates practical skills in data cleaning, SQL analysis, data modeling, DAX calculations, and dashboard development. The findings can support a better understanding of loan portfolio performance and help financial institutions identify potential credit risks.

Future Improvements

The project can be further improved by adding advanced risk analysis, loan default prediction models, and additional interactive filters to support more detailed analysis.